

Practice Questions for Mid Term Lab Exam of AI

Question # 01:

	1	2	3	4	5
A	S	1	#	2	3
B	2	#	2	#	1
C	3	2	1	2	2
D	#	2	#	1	3
E	2	1	2	2	G

Legend:

- S = Start
- G = Goal
- # = Obstacle
- Numbers = cost to move into that cell

Using the grid above, perform the following tasks:

1. BFS (Breadth-First Search):

- Find a path from S to G.
- Show the order in which nodes are expanded.

2. DFS (Depth-First Search):

- Find a path from S to G.
- Show the order in which nodes are expanded.

3. Uniform Cost Search (UCS):

- Use the cost values given in the grid for each move.
- Find the path from S to G with the minimum total cost.
- Show the path and total cost.

Question # 02:

	1	2	3	4	5
A	S	4	#	2	0
B	3	#	3	#	1
C	2	2	2	2	1
D	#	3	#	1	2
E	3	2	3	2	G

Legend:

- S = Start
- G = Goal
- # = Obstacle
- Numbers = heuristic value (h) — estimate of distance to G (for Greedy Best-First and A* Search)

Using the grid above, perform the following tasks:

1. Greedy Best-First Search:

- Use the heuristic values given to guide the search.
- Find a path from S to G.
- Show the order of nodes expanded and the path chosen.

2. A Search:*

- Use both path cost (assume moving to any adjacent cell costs 1) and the heuristic values.
- Find the optimal path from S to G.
- Show the order of nodes expanded, the path chosen, and the total cost.